

1. Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.

CODE:

```
[12]
✓ 1s # Artificial Neural Network using Backpropagation
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load Dataset
data = load_breast_cancer()
X = data.data
y = data.target

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Build ANN Model
model = MLPClassifier(hidden_layer_sizes=(10,),
                      activation='relu',
                      solver='adam',
                      max_iter=1000,
                      random_state=42)

# Train Model
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)
```

```
[12]
✓ 1s # Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

# Classification Report
print("\nClassification Report")
print(classification_report(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(6,5))
sns.heatmap(cm,
            annot=True,
            fmt='d',
            cmap='Blues',
            xticklabels=['Malignant', 'Benign'],
            yticklabels=['Malignant', 'Benign'])

plt.title("Confusion Matrix - ANN")
plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.show()
```

OUTPUT:

